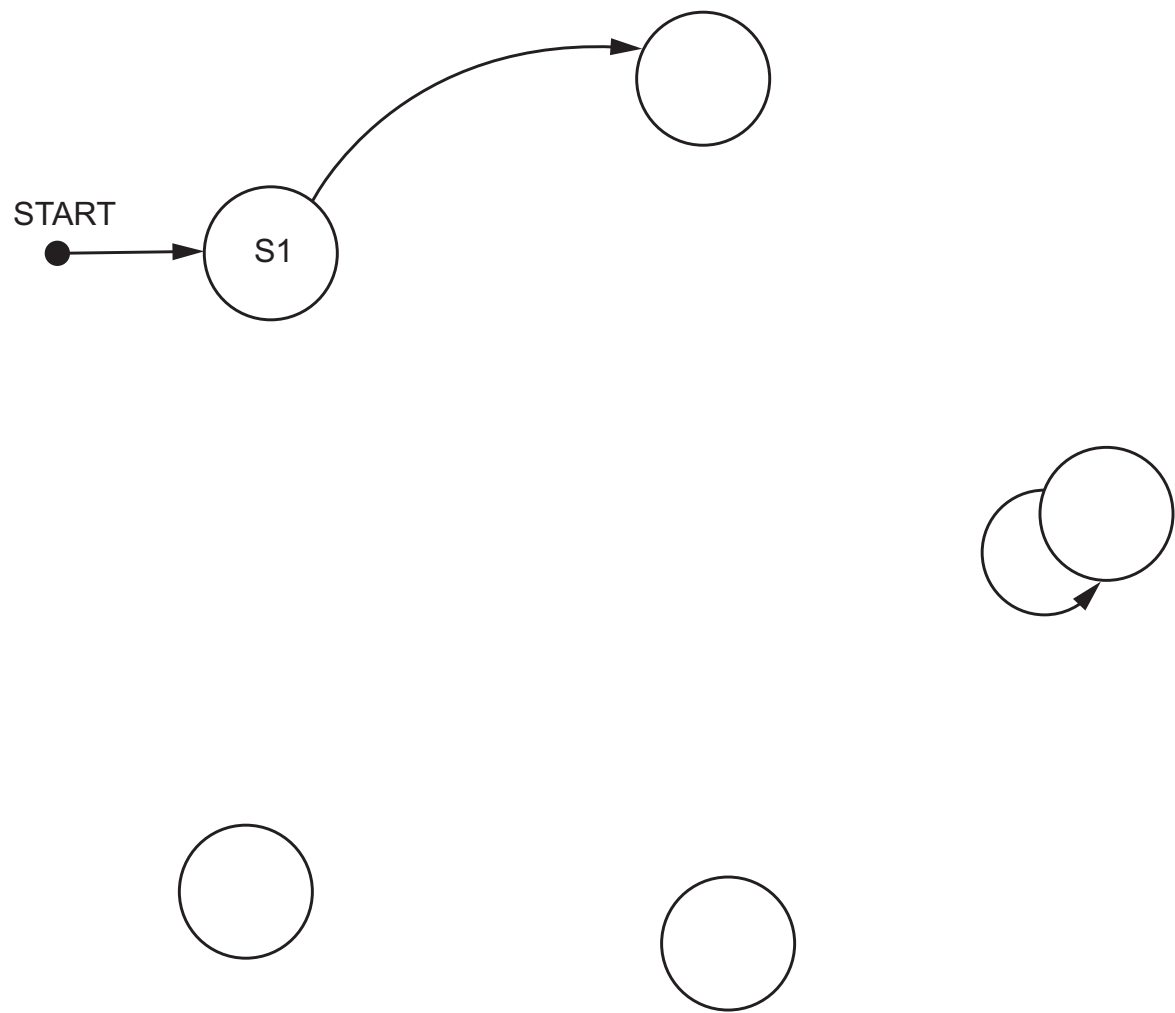


- 7
- There are several different ways to express an algorithm during the design of a program.
- (a)
- One part of the program contains an algorithm which is represented by a state-transition diagram.

The table shows the inputs, outputs and states for the algorithm:

Current state	Input	Output	Next state
S1	A1		S2
S2	A2	X4	S3
S3	A1	X1	S3
S3	A2	X1	S3
S3	A3	X3	S4
S3	A4		S2
S4	A3	X4	S5
S4	A4	X4	S5
S4	A1		S2

Complete the state-transition diagram to represent the information given in the table:



[5]

- (b)
- A structure chart is used to document a different part of the program, made up of five modules.

Program notes:

- module Setup calls either module Restart, or module Confirm
- module Confirm takes a string as a parameter and returns an integer
- module Modify has no parameters and returns a Boolean
- module Update takes a string as a parameter
- module Restart repeatedly calls module Update followed by module Modify
- module Restart takes a string as a parameter that is passed by reference.

Draw a structure chart to represent the relationship between the **five** modules, including all parameters and return values.

[4]